Saptarshi Mitra *May 14, 1995*

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Academic Details

Examination/Degree	University	Year	GPA/%
Ph.D.: (expected graduation: 2028) *EECS, Computer Engineering*	University of California, Irvine	2024-present	4.0/4.0
Master of Science: *Comm.& Electronics Engineering*	Technical University of Munich	July, 2021	1,5 (German Scale) First Class Distinction
Bachelor of Technology: *Electronics & Communication Engineering*	NIT Durgapur, India	May, 2017	9.14/10
Higher Secondary Examination	WBCHSE	May, 2013	88.8; Percentile : 99.97
10th Std. Examination	WBBSE	May, 2011	86.5

Selected Publications & Patents

- ***RaReCache: Bridging the Gap in Cross-Model KV Cache Reuse via Rank disagreement-based Selective Recomputation***. Sreetama Sarkar, **Saptarshi Mitra**, Sitao Huang, Souvik Kundu, Peter A. Beerel. (***arXiv***)
- ***ETA: Exit on Tool Arrival in Speculative Agent Loops***. **Saptarshi Mitra**, Xuan Zhu, Rohith Kukkala, Sitao Huang, Karthik Vijayan. *Machine Learning for Systems 2026 Workshop* (***NeurIPS'26***)
- ***SPADE: Floorplan-Aware Spatially Disaggregated Prefill And Decode Engines for Mixture-of-Agents LLM Inference on FPGAs***. Yifan Zhang, **Saptarshi Mitra**, Zhiheng Chen, Sitao Huang. (***Submitted***)
- ***Early Termination of AI Model Reasoning Using Speculative Tool Execution***. **Saptarshi Mitra**, Xuan Zhu, Rohith Kukkala, Karthik Vijayan. (***Patent Filed***)
- ***MOSAIC: Efficient Mixture-of-Agent Scheduling via Adaptive Aggregation and Inference Concurrency***. **Saptarshi Mitra**, Yifan Zhang, Rachid Karami, and Phyo Pyae Moe Aung, Nazmul Takbir, Sreetama Sarkar, Souvik Kundu, Sitao Huang. (***arXiv***)
- ***LowRank-SSM: Hardware-Software Co-Design for Rank-Reduced Mamba Acceleration on FPGA***. Haocheng Xu, Bhardwaj Bhat, Yu-an Chou, and Zhiheng Chen, Leyao Han, Yifan Zhang, Ye Qiao, **Saptarshi Mitra**, Sitao Huang. (***arXiv***)
- ***RAPID-LLM: Runtime Analysis of Parallel Infrastructure for Distributed LLM Training and Inference***. George Karfakis, Lime Yao, Binglu Chen, Faraz Tahmasebi, **Saptarshi Mitra**, Tianyue Pan, Hyoukjun Kwon, Puneet Gupta. (***arXiv***)
- ***Characterizing State Space Model and Hybrid Language Model Performance with Long Context***. **Saptarshi Mitra**, Rachid Karami, Haocheng Xu, Sitao Huang, Hyoukjun Kwon. *2026 IEEE International Symposium on Performance Analysis of Systems and Software* (***ISPASS'26***) [**Code**]
- ***Lookup Tables for Ultra Low-Bit Operations***. Saad Ashfaq, **Saptarshi Mitra**, Alexander Hoffman, Darshan Ganji,Ahmed Hassanien, Sudhakar Sah, Ehsan Saboori, MohammadHossein AskariHemmat. (***Patent Filed***)
- ***DeepliteRT: Computer Vision at the Edge*** . Saad Ashfaq, Alexander Hoffman, **Saptarshi Mitra**, Sudhakar Sah, MohammadHossein AskariHemmat, Ehsan Saboori. *The 34th British Machine Vision Conference* (***BMVC'23***)
- ***YOLOBench: Benchmarking Efficient Object Detectors on Embedded Systems***. Ivan Lazarevich, Matteo Grimaldi, Ravish Kumar, **Saptarshi Mitra**, Shahrukh Khan, Sudhakar Sah. *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops* (***ICCV'23***)
- ***DeepGEMM: Accelerated Ultra Low-Precision Inference on CPU Architectures using Lookup Tables***. Darshan C Ganji, Saad Ashfaq, Ehsan Saboori, Sudhakar Sah, **Saptarshi Mitra**, Mohammadhossein Askarihemmat, Alexander Hoffman, Ahmed Hassanien, Mathieu Leonardon. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops* (***CVPR'23***)
- ***Accelerating and Pruning CNNs for Semantic Segmentation on FPGA***. Pierpaolo Morì, Manoj-Rohit Vemparala, Nael Fasfous, **Saptarshi Mitra**, Sreetama Sarkar, Alexander Frickenstein, Lukas Frickenstein, Domenik Helms, Naveen Shankar Nagaraja, Walter Stechele, Claudio Passerone. *Proceedings of the 59th ACM/IEEE Design Automation Conference,*(***DAC'22***)
- ***Accelerating Semantic Image Segmentation on FPGA***. **Saptarshi Mitra**. *Technische Universität München'21*
- ***Arduino Based Foot Pressure Sensitive Smart Safety System for Industrial Robots***. Sayan Sarkar, Gautam Ghosh, Amitrakshar Mohanta, Atreye Ghosh, **Saptarshi Mitra**. *Second IEEE International Conference on Electrical,*

Computer and Communication Technologies (ICECCT'17)

Experience

Microsoft — Redmond, Washington
Research Intern, Applied Sciences Group — *June '26 – Sep '26*
- Speculative tool execution for on-device agentic LLM inference

Meta, Reality Labs — Redmond, Washington
ASIC Intern Contextual AI — *June '25 – Sep '25*
- On-device Contextual-AI for custom silicon in next-generation of XR-glasses.
- Hardware prototyping for Cascaded sensing to trigger high-power consuming event with lower-power consuming events.

Deeplite — Toronto, Canada
Embedded AI Engineer in Research & Development — *July '21 – May '24*
- Runtime for Neutrino model optimizer, providing significant speedup through extreme Quantization
- Automatic deployment of lightweight models on range of hardware platforms like embedded GPU, NPU, Raspberry Pi, MCU fostering hardware aware optimizing algorithms
- Bringup of vendor specific SBC and OS to run compressed models in custom runtime

Technical University of Munich — Munich, Germany
Student Research/Teaching Assistant — *Oct '20 – Feb '21*
- Automation of organization & grading of the Seminar course at the Chair of Design Automation
- Tutoring and mentoring graduate level students in VHDL System Design Lab (EI7403)

Intel — Munich, Germany
Internship & Working Student — *March '19 – July '20*
- During Internship, worked with Digital Design Verification team (Modem Technology Group) for development of a functional pre-silicon Verification IP (VIP) for a standardized AMBA debug interface.
- As a part of Intel Architecture, Graphics and Software (IAGS) group, validated system-level debug framework and various features related to CPU run control

Verizon — Hyderabad, India
Software Engineer — *Jun '17 – July '18*
Full stack developer (Angular & Spring boot) in Network & Technology Group

TCS Innovation Labs — Kolkata, India
Research Intern — *May '16 – July '16*
Localization of mobile Robots with ROS

Academic Projects

Accelerating Sparse Neural Networks for Semantic Image Segmentation on FPGA platforms *2020-2021*
- Novel dilated convolution irrespective of dilation rate and fitting the sparse weights in the on-chip buffer
- Efficient semantic image segmentation accelerator deploying state-of-the art model DeepLabV3+
- Hardware aware latency driven Genetic Algorithm based channel pruning for the custom accelerator

Sustaining Reduction in Carbon Emissions Based on Policy Decisions post COVID-19 Crisis *2020*
- Investigating the effects of stringent policy decisions imposed by various Governments on CO_2 emission
- Data collection, cleaning and building a web-application that implements a comprehensive data analysis pipeline with ML models in back-end and UI for interactive visualization

Radio-Controlled Multi-Functional Clock in Zync-7000 FPGA *2019-20*
- Specification and RTL design of the Global Finite State Machine module of the clock
- Integration with other modules like alarm clock, stopwatch and deployed on FPGA

Hardware-Oriented Implementation of IDEA (encryption algorithm) in Spartan-3e FPGA *2018-19*
- Used VHDL to implement and simulate the International Data Encryption Algorithm.
- Implemented Resource Constrained Scheduling for optimum resource utilization

Scholastic Achievement

Fellowship from UCI School of Engineering's Graduate Student Affairs office *2025-26*

Henry Samueli Endowed Fellowship, UCI EECS *2024-25*

Design Automation Conference (DAC) 2021 Young Fellow and best video award *Dec 2021*

Deutschlandstipendium, German National Scholarship Program for academic excellence, TUM *2018-21*

Scheme of Scholarship for College and University Student reg. of Government of India *2013*
Got an award on the basis of Higher Secondary Examination held by West Bengal Council of Higher Secondary Education

DST-INSPIRE Internship Science Camp *2011*
Selected and Participated at JBNSTS, Kolkata

Department of Science & Technology (DST) sponsored INSPIRE award *2010*
For academic excellence from 6th to 10th standard

'Best of Subject Category : Engineering' in India *2009*
At IRIS science fair conducted by Intel corporation, CII and DST held at Ahmedabad

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Skills

C, Python, VHDL, OpenCL, TVM, TFLite, ONNX Runtime, Verilog, SystemVerilog, UVM, VCS, Arduino, MATLAB, LaTeX, Logisim, Dash (Framework with Flask and React), Angular, C++, Java, Spring Boot, Assembly Level Programming, GDB, Keil uVision, HTML, CSS

Extra Curricular Activities

Semester Representative and Member of Quality Circle, MSCE, TUM *2018-2021*

Volunteer at TEDx Jadavpur University *2018*
played an active role in inviting speakers and arranging the event

Chief Events Coordinator of Prakriti - the Environmental club of NIT Durgapur *2016-17*
which was involved in activities like organizing plantation programs, observing earth hour, campus cleaning etc.

Google Online Marketing Challenge (GOMC) 2016 *2016*
Partnered with www.mayallyang.com on creating AdWords online marketing campaign for their expansion of Ecotourism . Ranked : Good

DuckDuckGo and Mozilla(l10n,e10s,webVR) *2014-Present*
Community member and Active Contributor

Volunteer at Blood Donation Camps organized by State Blood Transfusion Council, West Bengal *2014*